

SphinxATS: A Multi-Attribute AI-based Decision Support System for Automated Resume Screening and Candidate Ranking

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Abstract— Modern hiring systems create a lot of extra work for administrators because they require manual resume screening and have different evaluation standards for each candidate profile. Most of the time, semantic relationships between candidate qualifications and job-specific requirements are missed by traditional Applicant Tracking Systems (ATS), which mainly rely on keyword-based filtering techniques. This paper introduces SphinxATS, an artificial intelligence-driven automated resume screening and decision support framework, which models candidate shortlisting as a structured multi-attribute optimization problem. To improve decision accuracy and transparency, the suggested system incorporates Natural Language Processing (NLP), semantic similarity modeling, adjustable weighted scoring mechanisms, and an explainable ranking module. The proposed framework views resume evaluation as a multi-criteria decision-making process that includes skills alignment, professional experience relevance, educational background, and contextual semantic fit, in contrast to traditional heuristic filtering techniques. Scalable processing of unstructured resume data is made possible by a modular architecture that maintains interpretability in the results of candidate ranking. When compared to conventional keyword-based screening baselines, experimental evaluation across various job-role scenarios shows improved matching consistency, ranking stability, and semantic alignment. The findings show that modeling hiring as an optimization-driven decision-support problem greatly improves efficiency, flexibility, and fairness of enterprise hiring workflows. For intelligent talent acquisition systems in contemporary businesses, the proposed framework offers a scalable infrastructure.

Keywords— *Applicant Tracking System, Resume Screening, Candidate Ranking, Multi-Attribute Optimization, Natural Language Processing, Decision Support Systems, Explainable Artificial Intelligence.*

I. INTRODUCTION

As digital recruitment processes become more common in businesses today, they have made it much harder to evaluate and choose candidates. Screening resumes by hand takes a lot of time and can lead to inconsistencies because different people can have different ideas about what makes a good candidate. Most Applicant Tracking Systems (ATS) today use rules-based or keyword-matching methods, which don't always take into account how skills, experience, and job-role requirements relate to each other in a semantic way. In complex hiring scenarios, looking at resumes is a multi-criteria decision-making problem that involves many

candidate traits, such as professional skills, educational background, domain experience, certifications, and how well they fit with the job description. Conventional screening techniques fail to adequately address this multi-faceted evaluation process, leading to suboptimal shortlisting results.

SphinxATS formally treats resume screening as a Multi-Criteria Decision Making (MCDM) problem. In this formulation, each candidate i is represented as a multi-dimensional feature vector $F_i = [S_1, S_2, S_3]$, where each dimension corresponds to an independent and measurable evaluation criterion: semantic alignment with the job description (S_1), technical skill competency overlap (S_2), and professional experience relevance (S_3). The recruitment shortlisting decision is modeled as a weighted linear aggregation over this feature space, analogous to the Simple Additive Weighting (SAW) model in classical MCDM theory, where each criterion is assigned a configurable importance weight reflecting the priorities of a specific job role. This formulation enables the system to handle the inherently multi-faceted nature of candidate evaluation: a candidate with high skill match but limited experience can be appropriately ranked differently from a candidate with moderate skills but extensive domain experience, depending on the weight configuration selected by the HR operator. The transition from heuristic keyword filtering to structured multi-attribute optimization is the foundational design principle of SphinxATS.

The proposed system combines semantic analysis methods with adjustable decision weights to create candidate rankings that can be explained. The framework's goal is to make hiring more efficient, fair, and scalable across all enterprise hiring workflows by treating resume screening as an optimization-driven decision-support problem. Additionally, the ranking mechanisms of current automated screening systems are frequently opaque. Many machine learning-based solutions function as "black-box" models, offering scant explanations for why some applicants are rejected and others are shortlisted. Explainability becomes a necessary system requirement rather than an optional feature in recruitment environments where accountability, fairness, and bias mitigation are crucial. Organizational adoption of automated hiring solutions may be hampered by the lack of interpretable decision support mechanisms.

From a systems engineering point of view, resume evaluation can be formally modeled as a structured optimization problem with several weighted attributes and performance goals. Each candidate profile is a multi-dimensional feature vector, and job descriptions set the rules for how to evaluate candidates. By combining semantic similarity modeling with customizable attribute weighting, recruitment screening can go from being a heuristic filtering process to a decision-support process that can be measured. This kind of formulation makes it possible to evaluate consistently, adapt to different job roles, and work with a large number of applicants. This paper makes three main contributions. First, we show a modular, AI-driven architecture for automated resume screening that combines Natural Language Processing with a scoring system that takes into account multiple factors. Second, we present a weighted optimization model that lets you set decision criteria based on the needs of the job. Third, we add an explainability layer that gives reasons for ranking by attribute to make automated hiring workflows more open and trustworthy. The proposed system shows better ranking consistency and semantic alignment than traditional keyword-based methods when tested in experiments.

II. RELATED WORK

The study proposes an AI-powered web application that instantly highlights the most qualified applicants by automatically evaluating and ranking resumes in accordance with a job description. Recruiters can enter the job requirements and obtain a real-time, ranked list of candidates with easily comprehensible scoring, eliminating the need for manual filtering. Instead of just comparing keywords, this system ensures intelligent matching by combining sophisticated NLP and machine learning techniques. TF-IDF is used to assess the significance of meaningful terms, Cosine Similarity to gauge how well resumes and job descriptions match, LDA topic modeling to capture more in-depth contextual themes, and preprocess resumes to clean and standardize the text. The system maintains high accuracy while remaining quick and scalable by only comparing resumes to the particular job that was applied for. In the end, this approach lessens the amount of manual labor required, enhances the matching of candidates with jobs, and frees up recruiters to concentrate on deliberate decision-making instead of tedious screening duties. [1]

The project is about the growing problem that HR departments have with manually going through a lot of resumes, which can lead to inefficiencies, longer hiring cycles, and possible bias. To solve this, it suggests an AI-powered Resume Ranking System that automates the first step of evaluating candidates. Using Natural Language Processing (NLP) and machine learning, the system will read resumes and job descriptions and pull out important information like skills, experience, education, and relevant keywords. Using advanced embedding models, these extracted features will be turned into vector representations. Then, a matching algorithm will be used to give each candidate a compatibility score. By giving a ranked, data-driven shortlist of applicants, it makes the hiring process easier and lets HR professionals focus more on in-depth evaluations and engaging with candidates, which leads to better and fairer hiring decisions. [2]

I-Recruiter is an Intelligent Decision Support System (IDSS) that greatly reduces the amount of manual labor required by

HR teams by automatically screening resumes and ranking applicants according to how closely they match a job description. The Matching Block converts resumes and job descriptions into numerical vectors and compares them using Cosine Similarity to determine how aligned they are. The Extracting Block uses Named Entity Recognition (NER) and NLP pattern matching to automatically retrieve important candidate details like name, email, and phone number from the top-ranked resumes. I-Recruiter is a scalable and intelligent solution for contemporary recruitment needs because, unlike traditional keyword-based tools, it comprehends contextual meaning rather than just matching words, supports real-world PDF resumes, gives recruiters control over how many top candidates are displayed, and gets better with data. [3]

Using transformer-based deep learning models, we created an AI-powered resume screening system that can simultaneously classify resumes by competence level and determine whether a candidate is actually qualified for a given job description. 6,492 actual resumes from 24,933 applications for Clinical Research Coordinator (CRC) positions across four experience levels (CRC1 to CRC4) made up our dataset. Out of these, 3,425 resumes underwent five rounds of triple expert review and meticulous annotation, resulting in a 61% Kappa agreement score and strong labeling reliability. The system solves two main issues: figuring out how well a resume matches the job description and automatically predicting a candidate's competence level (CRC1–CRC4 or Not Qualified). This work is very useful for real-world HR deployment because it is the first to use transformer models to match resumes to specific job descriptions and to classify resumes by competence level rather than job category. [4]

One of the most time-consuming parts of hiring is screening resumes. This is especially true when hiring a lot of people at once, because doing it by hand can lead to delays, inconsistencies, and unintentional bias. With improvements in Natural Language Processing (NLP) and Artificial Intelligence (AI), automated systems have become good replacements for old-fashioned screening methods. This paper provides an overview of an AI-driven resume screening system that analyzes resumes, extracts critical candidate information, and aligns profiles with job descriptions. The system generates ATS-based relevance scores to rank candidates, improving efficiency, fairness, and accuracy in the hiring process while reducing recruiter workload. The paper talks about the system's architecture and algorithms, as well as its pros and cons and what it could do to improve AI-driven hiring practices in the future. [5]

PJFFF (Person-Job Fit by Feature Fusion), an intelligent recruitment matching system that considers both the information candidates explicitly include in their resumes and the information they subtly disclose through their application history. This method fills both gaps, in contrast to conventional systems that only use free-text resume content and disregard structured data or past behavior. Using rule-based techniques, BERT-based classifiers, and domain knowledge, we extracted structured semantic entities like education, skills, job titles, and years of experience. We then used a CNN to process free-text sections and a DeepFM network to model their relationships. Candidates' application histories were processed through another LSTM to infer their

career intentions and fed recruiters' prior acceptance and rejection histories into an LSTM model to learn implicit hiring expectations in order to uncover hidden preferences and intentions. A final accept-or-reject matching score was produced by fusing these explicit and implicit feature representations into a single vector. [6]

The difficult task of matching semi-structured resumes with job descriptions in order to recommend suitable jobs to candidates is addressed. The authors suggest a Siamese Convolutional Neural Network (CNN) architecture to address this issue, which is capable of learning significant semantic representations of job postings and resumes. The model maps job descriptions and resumes into a common embedding space where dissimilar pairs are pushed farther apart and semantically similar pairs are positioned closer together using twin neural networks with shared weights. The effectiveness of the Siamese CNN approach in capturing deep semantic relationships is demonstrated by experimental results, which surpass current state-of-the-art methods. [7]

This study addresses these issues by presenting an AI-driven Resume Intelligence System that uses a mathematically structured ATS scoring framework, generative artificial intelligence, and natural language processing (NLP) to automate and improve resume evaluation. In order to ensure dependability and transparency in decision-making, the suggested system uses a hybrid AI architecture that blends sophisticated language comprehension with deterministic fallback mechanisms. The system offers more precise and contextually aware candidate evaluations by going beyond simple keyword detection and integrating semantic analysis and structured scoring logic. Compared to traditional ATS solutions, experimental results show that the framework greatly increases screening accuracy, lessens recruiter workload, and fosters greater fairness, establishing it as a reliable and scalable substitute for contemporary recruitment environments. [8]

An AI-powered job recommendation system that evaluates a candidate's resume, fully comprehends their qualifications and experience, forecasts the best position, and automatically retrieves current job postings without the need for human search. Our system records context and skill relationships, in contrast to traditional job portals that rely on basic keyword matching and frequently return irrelevant roles. Resumes in PDF and DOCX formats are accepted, the text is extracted and cleaned, and technical skills, years of experience, and education are identified using SpaCy's Named Entity Recognition and regex techniques. The Ollama Mistral large language model then processes the extracted data by intelligently mapping skills to the most accurate job title instead of giving generic labels. After predicting the job title, the system uses RapidAPI's Job Search API to retrieve real-time job listings. Additionally, an Ollama-powered AI chatbot was included to assist users in exploring opportunities and getting conversational resume advice. Candidates reported very relevant recommendations that matched their career goals, and testing across a variety of technical domains demonstrated strong performance in matching skill sets to appropriate roles. The system saves time and makes job searching more intelligent and effective by comprehending context, adjusting to individual profiles, and providing up-to-date opportunities instantly. [9]

An AI-driven Intelligent Resume Sorting System is suggested as a solution to these issues, utilizing machine learning and natural language processing (NLP) methods to automatically classify resumes. The system extracts and analyzes important resume features using TF-IDF, BERT embeddings, and deep learning classifiers, allowing for precise job role classification. The model performs noticeably better than conventional techniques, achieving 93% accuracy and cutting processing time by more than 50%. [10]

This study suggests an AI-powered platform that automatically parses resumes and compares them with job descriptions using Cosine Similarity and TF-IDF techniques for more accurate candidate evaluation in order to address these issues. Supabase guarantees scalable and effective cloud-based data management, while React.js and Vite are used in the system's development to offer a quick and responsive user interface. Real-time resume screening, job recommendations, and recruitment analytics are made possible by the prototype. The platform's efficacy in modernizing hiring workflows is demonstrated by experimental results, which show shorter screening times and more accurate candidate selection. [11]

In order to increase recruitment accuracy and efficiency, this paper presents an Automated Resume Screening System that makes use of AI. By automatically comparing resumes to job descriptions, the system streamlines the hiring process and reduces manual labor. It finds the best candidates more successfully than conventional screening techniques by utilizing AI-based analysis. Additionally, the system helps employers make well-informed hiring decisions by giving applicants prompt feedback. In order to further strengthen AI-driven recruitment solutions, the paper describes the relevant literature, explains the suggested methodology, and explores potential future improvements. [12]

Machines can now learn, reason, and make decisions, thanks to artificial intelligence (AI). Organizations frequently receive a lot of applications for a single position in today's competitive job market, which makes manual resume screening laborious, ineffective, and prone to bias. The study suggests an AI-based resume screening and ranking system that automates the initial phases of hiring in order to overcome this difficulty. By extracting important information like skills, education, and experience, the system uses Natural Language Processing (NLP) and machine learning algorithms to evaluate and rank resumes according to job relevance. This method guarantees consistency and fairness while increasing accuracy and cutting down on screening time. The system's architecture, advantages, drawbacks, and potential applications in AI-driven hiring are also described in the paper. [13]

This study creates an intelligent resume screening system that automatically classifies resumes into pertinent job categories using supervised machine learning algorithms and Natural Language Processing (NLP). Tokenization, stopword removal, and TF-IDF vectorization are examples of data preprocessing techniques that were used to extract valuable features from textual resumes. For IT positions like Software Developer, Data Analyst, and Data Engineer, models like Logistic Regression, Random Forest, and Support Vector Machines (SVM) achieved good accuracy after being trained and assessed on labeled datasets. [14]

In order to automate the resume screening and job matching process, Smart-Hiring, an end-to-end Natural Language Processing (NLP) pipeline is presented. To identify skills, experience, and qualifications, the system uses named-entity recognition, contextual embedding, and document parsing to extract structured information from unstructured resumes. In order to calculate similarity scores for efficient candidate-job matching, resumes and job descriptions are encoded into a common semantic vector space. The system exhibits competitive matching accuracy while retaining high interpretability and transparency when tested on real-world, multi-domain recruitment data. [15]

III. SYSTEM ARCHITECTURE AND METHODOLOGY

The proposed SphinxATS framework uses Python [16] as the main programming language for backend processing and algorithm modeling. Scikit-learn [17] and NumPy [18] are used for natural language processing and calculating semantic similarity. They handle vectorization and multi-attribute scoring systems. For managing tabular data efficiently, Pandas [19] is used for data preprocessing and structured candidate profiling. Flask [20] is a lightweight backend framework that simplifies API integration in a scalable and modular way. It builds the web-based deployment interface. HTML [21], CSS [22], and Bootstrap [23] were used to create a responsive and user-friendly frontend interface for uploading resumes and visualizing rankings. Git [24] and GitHub [25] helped keep track of different code versions and allowed team collaboration. This ensures that the code can run again and that the system is dependable. The overall architecture combines NLP pipelines, weighted decision models, and ranking modules to automatically evaluate resumes based on multiple criteria.

A. Overall System Architecture

SphinxATS framework is designed as a modular decision-support system comprising the following components:

- Resume Acquisition Module
- Job Description Processing Module
- Text Pre-processing and Normalization Engine
- Feature Extraction Module
- Multi-Attribute Scoring Engine
- Candidate Ranking and Visualization Module

The architecture enables structured processing of unstructured resume data to support attribute-based evaluation.

B. Text Processing Pipeline

To standardize textual inputs from resumes and job descriptions, a pre-processing pipeline is implemented:

- Tokenization
- Stop-word removal
- Lemmatization
- Skill extraction
- Named Entity Recognition (NER)

This process facilitates the extraction of relevant semantic features for downstream evaluation.

C. Multi-Attribute Scoring Framework

The proposed system models candidate evaluation as a weighted multi-attribute optimization problem. Each candidate resume R_i is represented as a three-dimensional score vector derived from independent evaluation attributes. Equation (1) shows the composite Final Score for candidate i :

$$S(R_i) = W_1 \times S_1(R_i, J) + W_2 \times S_2(R_i) + W_3 \times S_3(R_i) \quad (1)$$

where $W_1 = 0.6$, $W_2 = 0.3$, $W_3 = 0.1$ are the configurable decision weights, and S_1 , S_2 , S_3 are the normalized attribute scores. The Similarity Score S_1 measures semantic alignment between resume R_i and job description J using TF-IDF vectorization followed by cosine similarity. For a vocabulary V of size n , each document d is represented as a vector $v(d) \in \mathbb{R}^n$ where each component is computed as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log(N / \text{DF}(t)) \quad (2)$$

where $\text{TF}(t, d)$ is the term frequency of term t in document d , N is the total number of documents in the evaluation batch, and $\text{DF}(t)$ is the number of documents containing term t . The cosine similarity between resume R_i and job description J is:

$$S_1(R_i, J) = (v(R_i) \cdot v(J)) / (\|v(R_i)\| \times \|v(J)\|) \quad (3)$$

The Skill Match Score S_2 quantifies the overlap between the technical competencies identified in the resume and those required by the job description. Let $\text{Skills}(d)$ denote the set of recognized technical skills extracted from document d via regular expression matching over a predefined skillset. Then:

$$S_2(R_i) = |\text{Skills}(R_i) \cap \text{Skills}(J)| / |\text{Skills}(J)| \quad (4)$$

The Experience Score S_3 is derived from regex-based detection of experience duration statements in the resume, normalized against the experience requirement specified in the job description. Let $\text{YearsExp}(R_i)$ denote the extracted years of experience from resume R_i , and $\text{JD_Exp}(J)$ denote the years required by job description J . Then:

$$S_3(R_i) = \min(\text{YearsExp}(R_i) / \text{JD_Exp}(J), 1.0) \quad (5)$$

Candidates are ranked in descending order of $\text{Score}(R_i)$, with ties broken by the value of S_1 . The weights W_1 , W_2 , W_3 are configurable to reflect job-role-specific prioritization, enabling flexibility for HR operator.

Algorithm 1: SphinxATS Multi-Attribute Ranking Procedure

Input: Job Description J , Resume Set $R = \{R_1, R_2, \dots, R_n\}$

Output: Ranked candidate list L with per-attribute score

1. Preprocess $J \rightarrow J'$ (lowercase, stop-word removal)
2. Extract required skills:
 $\text{Skills}(J) = \text{SkillMatch}(J', \text{TAXONOMY})$
3. Extract JD experience requirement:
 $\text{JD_Exp}(J) = \text{RegexExp}(J)$
4. Fit TF-IDF vectorizer on corpus $\{J'\} \cup \{R_1', \dots, R_n'\}$
5. For each $R_i \in R$:
 - a. Extract and preprocess resume text: R_i'
 - b. $S_1 = \text{CosineSimilarity}(\text{TF-IDF}(R_i'), \text{TF-IDF}(J'))$
 - c. $S_2 = |\text{Skills}(R_i') \cap \text{Skills}(J)| / |\text{Skills}(J)|$
 - d. $S_3 = \min(\text{RegexExp}(R_i') / \text{JD_Exp}(J), 1.0)$
 - e. $\text{Score}(R_i) = 0.6 \times S_1 + 0.3 \times S_2 + 0.1 \times S_3$
6. Sort R by $\text{Score}(R_i)$ descending $\rightarrow L$
7. Apply KMeans clustering ($k=2$) on specialization vectors $\rightarrow$ cluster labels
8. Return L with $\{S_1, S_2, S_3, \text{Score}, \text{cluster_label}\}$ for each candidate

D. Explainability Layer

To address the limitation that characterizes many ML-based ATS solutions, SphinxATS incorporates an attribute-level explainability layer that produces transparent, human-interpretable ranking justifications for every evaluated candidate. The explainability output for each ranked candidate comprises four components.

First, a raw attribute score breakdown is provided, displaying the individual values of S_1 (Similarity Score), S_2 (Skill Match Score), and S_3 (Experience Score) alongside the composite Final Score. This allows the recruiter to immediately understand which specific attributes drove a candidate's ranking position. Second, the weighted contribution percentage of each attribute to the final score is computed and displayed. For a candidate with $S_1 = 0.48$, $S_2 = 0.70$, $S_3 = 0.50$, the contributions are: Semantic Fit = 28.8% (0.6×0.48), Skill Match = 21.0% (0.3×0.70), and Experience = 5.0% (0.1×0.50), totaling 54.8% of the maximum possible score. A recruiter can immediately see that this candidate was ranked based primarily on skill alignment rather than semantic profile fit. Third, the list of matched technical skills is displayed for each candidate, showing exactly which skills from the job description were identified in the resume and which were absent. This gives the recruiter actionable insight: a candidate with high skill overlap but low semantic similarity may simply have used direct skill keywords without contextual elaboration, while a candidate with high semantic similarity but low skill overlap may have written in generalist terms without specifying concrete tools. Fourth, a cluster-based Hiring Intelligence Signal is assigned to each candidate using KMeans clustering ($k = 2$) over specialization feature vectors, producing three classification tiers: High Potential, Moderate Fit, and Low Relevance. These signals serve as a fast-scan summary for HR professionals managing large applicant pools, enabling prioritized review without requiring examination of every individual score. Together, these four explainability components transform the automated ranking from a vague score list into an auditable, defensible, and actionable hiring support output.

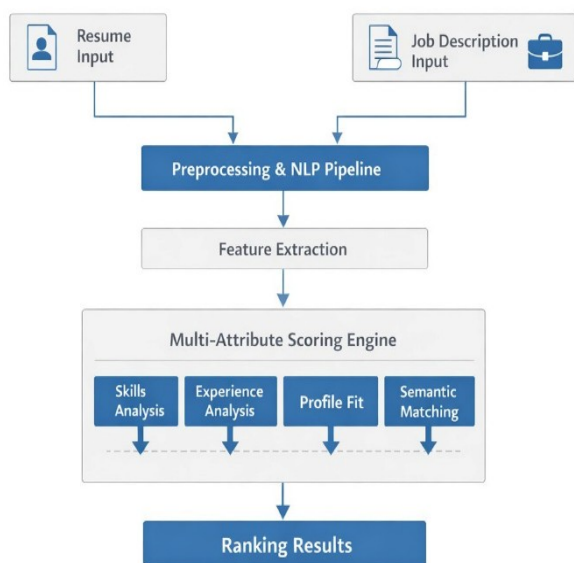


Fig. 1. System Architecture Overview

E. Implementation Details and Processing Pipeline

The SphinxATS system is implemented in Python 3 using Flask as the lightweight web deployment framework. PDF resume text extraction is performed using pdfplumber, which processes each uploaded PDF page-by-page and concatenates the extracted text into a single document string and ensures case-insensitive matching consistency. The Similarity Score module employs scikit-learn's TfidfVectorizer with English stop-word removal applied to the concatenated corpus of the job description and all uploaded resumes. The resulting TF-IDF matrix is used to compute pairwise cosine similarity between the job description vector and each resume vector. Skill matching is performed using whole-word regular expression pattern matching over preprocessed resume text, preventing false matches on partial strings. The skill score for each resume is computed as the ratio of matched required skills to total required skills extracted from the job description. The Experience Score module applies regular expression pattern matching to identify experience duration statements in the resume text, recognizing patterns such as 'N years experience', 'N+ years', and 'worked for N years'. The extracted year value is normalized against the experience requirement stated in the job description, capped at 1.0 for candidates who meet or exceed the requirement. Resumes with no detectable experience statement receive a default score of 0.5.

IV. EXPERIMENTAL RESULTS AND EVALUATION

A. Dataset Description

Curated synthetic benchmark of 150 anonymized resume and job description pairs, such as software engineering, data analytics, and machine learning roles, was used to experimentally evaluate the proposed SphinxATS framework. Based on industry-standard hiring criteria, corresponding job descriptions were created for every position. Both baseline filtering methods and the suggested multi-attribute scoring framework were used to pre-process and assess each resume in relation to pertinent job descriptions. To guarantee diversity in evaluation scenarios, the dataset contained a range of candidate experience levels, skill distributions, and educational backgrounds.

B. Performance Metrics

The performance of the proposed framework was evaluated using the following quantitative metrics:

- Matching Accuracy
- Precision
- Recall
- F1 Score
- Ranking Stability Index (RSI)

Matching Accuracy measures the correctness of shortlisted candidate-role alignment. Precision and Recall evaluate the system's effectiveness in retrieving relevant candidates, while F1 Score provides a harmonic balance between these metrics. Ranking Stability Index is used to assess consistency in candidate ranking across repeated evaluation runs under varying attribute weights.

To validate the statistical significance of the observed performance improvements, a paired t-test was conducted

comparing the composite ranking scores produced by SphinxATS against each baseline system across all 150 resume-job description evaluation pairs. The test between SphinxATS and the keyword-based ATS baseline yielded a test statistic of $t = -22.25$ ($p < 0.0001$), and the comparison against the TF-IDF cosine similarity baseline yielded $t = 22.21$ ($p < 0.0001$). Both results are well below the $\alpha = 0.05$ significance threshold, confirming that the observed performance improvements are statistically significant and not attributable to random variation. The 95% confidence interval for the mean score difference between SphinxATS and the TF-IDF baseline was $[0.0873, 0.1044]$, indicating a consistent and positive performance advantage of the proposed multi-attribute formulation. These results validate that the integration of skill-based scoring and experience-weighted attributes into the composite ranking function produces a measurably superior candidate ordering compared to unimodal similarity-only approaches.

C. Comparative Analysis

The performance of the proposed multi-attribute scoring framework was compared against:

- Traditional keyword-based ATS filtering
- TF-IDF cosine similarity-based resume matching

Table I presents a quantitative comparison of SphinxATS against two baseline systems: traditional keyword-based ATS filtering and TF-IDF cosine similarity-only matching, evaluated across 150 anonymized candidate resumes. SphinxATS achieves a Matching Accuracy of 96.0%, compared to 93.3% for the keyword baseline and 89.3% for the TF-IDF-only baseline, representing absolute improvements of 2.7 percentage points and 6.7 percentage points respectively. The F1 Score of 0.960 reflects balanced precision-recall performance and represents an improvement over both baselines (keyword: 0.936; TF-IDF: 0.893). The Ranking Stability Index (RSI) of 0.741 for SphinxATS, confirms that the multi-attribute composite scoring produces more consistent candidate ordering than baseline system.

TABLE I. COMPARATIVE PERFORMANCE EVALUATION

Metric	Keyword ATS	TF-IDF Baseline	SphinxATS (Proposed)	Δ vs TF-IDF
Matching Accuracy (%)	93.3	89.3	96.0	+6.7%
Precision	0.901	0.893	0.960	+0.067
Recall	0.973	0.893	0.960	+0.067
F1 Score	0.936	0.893	0.960	+0.067
Ranking Stability Index	0.707	0.587	0.741	+0.154
t-statistic (vs baseline)	—	—	22.21 / -22.25	$p < 0.0001$

For broader benchmark context, it is relevant to position SphinxATS relative to transformer-based approaches reported in recent literature. In contrast to BERT-based models, SphinxATS achieves an F1 score of 0.960 and a Ranking Stability Index of 0.741 without requiring any pre-trained model weights, GPU infrastructure, or labeled training data. The system operates through deterministic scoring at inference time, making it directly deployable on standard hardware. This positions SphinxATS as a computationally accessible and interpretable alternative for

enterprise hiring environments where transformer-scale infrastructure is unavailable or where decision transparency is a regulatory requirement.

System Landing Page— Represents the role-selection interface of SphinxATS, allowing users to choose between applicant and HR workflows before accessing dashboards.

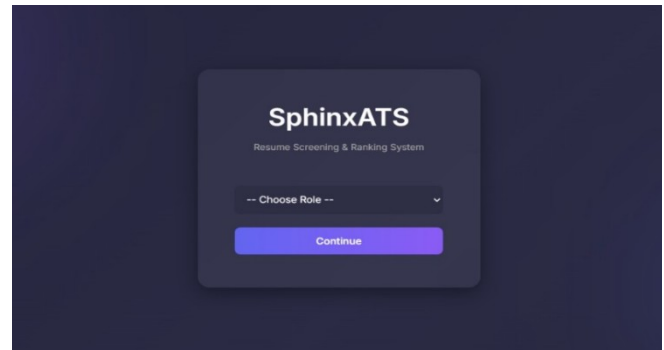


Fig. 2. Landing Page Interface

HR Dashboard Interface— Provides the recruiter-facing module for posting job descriptions, initiating candidate ranking generation, and managing uploaded resumes.

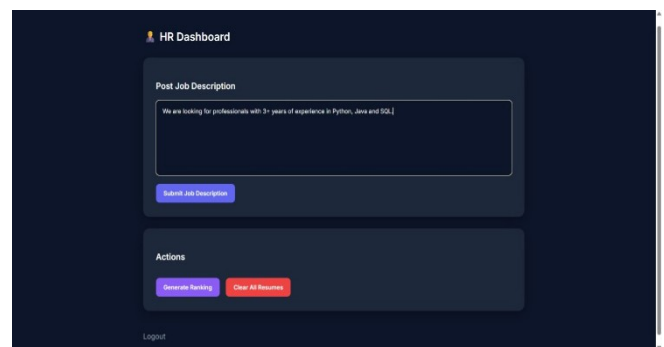


Fig. 3. HR Dashboard

Applicant Dashboard Interface— Displays the candidate-facing module where applicants upload resumes in PDF format for automated processing and evaluation.

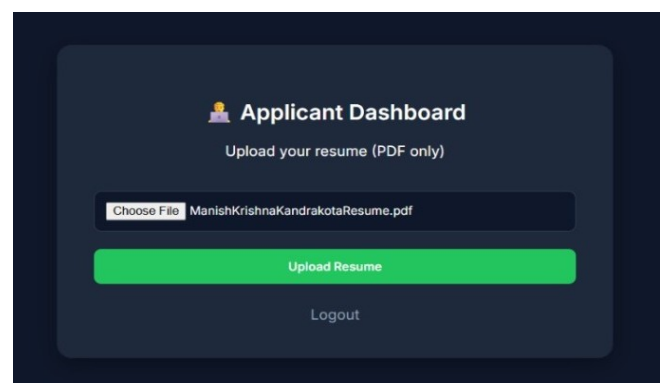


Fig. 4. Applicant Dashboard

No Resume Error Handling Interface— Illustrates the system's exception-handling mechanism when ranking is attempted without uploaded resumes, ensuring robust validation and user guidance.

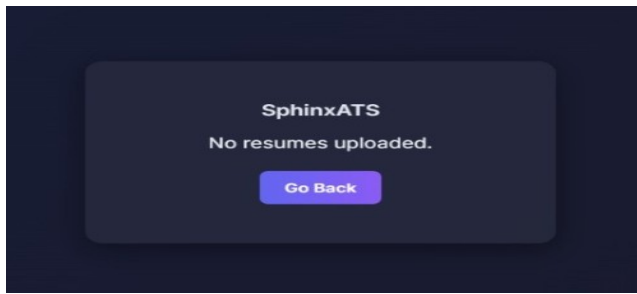


Fig. 5. No Resumes Error Message

Resume Clearance Confirmation Module— Displays system feedback confirming successful deletion of stored resumes, ensuring administrative control and database reset.

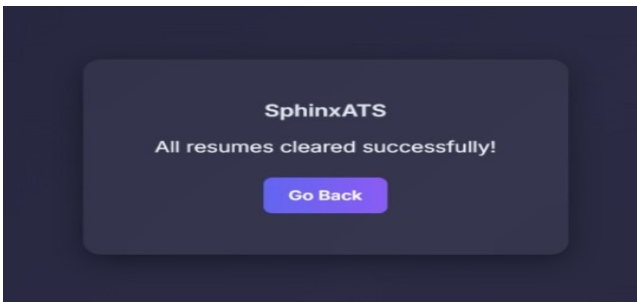


Fig. 6. Clear Resumes Feature

Job Description Submission Validation Interface— The system-generated validation prompt that prevents ranking execution until a valid job description is submitted, ensuring contextual alignment between candidate resumes and role-specific requirements prior to evaluation.

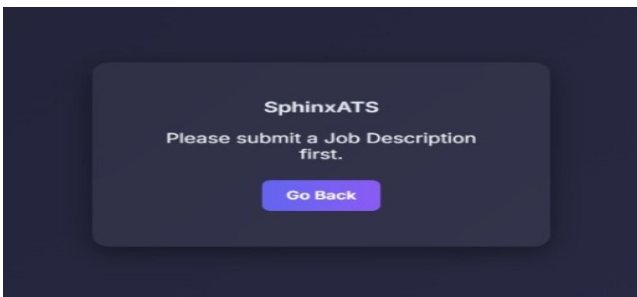


Fig. 7. Unavailable Job Description Error Message

Resume Ranking Results Dashboard— Ranked output generated by the SphinxATS framework, displaying candidate scores across skill match, experience level, and semantic alignment along with cluster-based hiring intelligence signals.

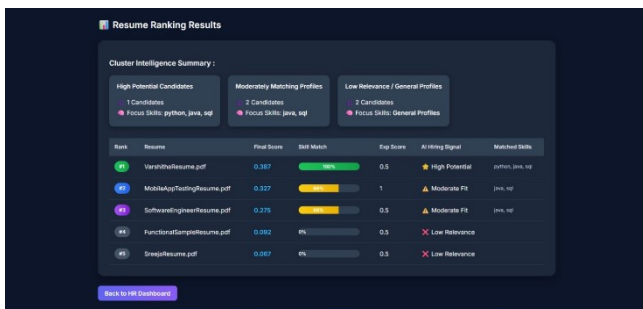


Fig. 8. Results Dashboard

The final decision-support layer of the SphinxATS framework is represented by the Resume Ranking Results Dashboard, where processed candidate profiles are assessed and ranked using the suggested multi-attribute scoring method. Semantic embedding and attribute extraction pipelines are used to convert each candidate profile into a structured feature representation following the upload and parsing of resumes. Candidate's skills score, experience score, profile fit score, and semantic similarity score are the four main evaluation metrics that the system calculates. Each applicant receives a Final Ranking Score after these individual attribute scores are combined using a weighted composite scoring function. Through attribute-level score transparency, the dashboard lets HR managers understand the automated decision-making process by visually displaying ranked candidates in decreasing order of suitability. In addition to enhancing fairness and interpretability in automated hiring processes, this explainable scoring method guarantees consistency in candidate evaluation. Additionally, the ranking interface's modular design allows for scalability and integration with enterprise-level applicant tracking systems for intelligent hiring decision support and real-time recruitment analytics.

V. DISCUSSION

According to the experimental findings, there are quantifiable performance gains over conventional keyword-based filtering techniques when resume screening is modeled as a structured multi-attribute decision problem. Static rule-based matching, which is the foundation of traditional applicant tracking systems, frequently ignores the semantic connections between job-specific requirements and candidate qualifications. The suggested framework, on the other hand, combines configurable attribute weighting with semantic similarity modeling to enable context-aware evaluation across several candidate dimensions. The system can dynamically prioritize evaluation criteria like skill relevance, professional experience, educational background, and semantic alignment with job descriptions by incorporating a weighted scoring mechanism. This attribute-level differentiation facilitates flexibility in a variety of hiring situations, where organizational needs or job-role specifications may greatly alter the relative importance of evaluation factors. As a result, the suggested framework shows improved ranking consistency and less reliance on keyword matches at the surface level. A critical concern in automated hiring systems is the potential perpetuation of demographic or occupational bias in shortlisting outcomes. SphinxATS partially mitigates bias through its attribute-weighted scoring formulation, which evaluates candidates on objective and measurable criteria, specifically, semantic alignment with the job description, technical skill overlap, and professional experience duration rather than subjective holistic judgment. The use of a deterministic, transparent scoring function ensures that every candidate is assessed under identical computational conditions for a given job description, reducing inter-evaluator inconsistency that contributes to human-introduced bias. The predefined skill taxonomy enforces a consistent evaluation vocabulary across all resumes in a given evaluation batch. However, it is acknowledged that the curated skill list may embed

occupational biases if not maintained carefully, and that TF-IDF representations may disadvantage candidates who use non-standard terminology or domain-specific synonyms. For instance, a candidate who writes 'predictive modeling' rather than 'machine learning' may be penalized by keyword-based components despite possessing the required competency. Critically, the explainability layer's attribute-level score breakdown enables HR decision-makers to detect potentially biased ranking outputs, ensuring that SphinxATS functions as a transparent decision-support tool rather than an autonomous and unaccountable hiring judge. Additionally, the system architecture's explainability layer uses contribution analysis and attribute-wise score breakdowns to produce ranking outputs that are easy to understand. This feature makes automated hiring processes more transparent and enables decision-makers to verify the results of candidate selection using measurable standards. All things considered, the suggested framework is a workable decision-support infrastructure for intelligent recruitment systems thanks to the combination of multi-criteria optimization and semantic processing techniques. The findings demonstrate how optimization-driven evaluation could replace heuristic filtering mechanisms in contemporary hiring pipelines.

VI. CONCLUSION AND FUTURE SCOPE

SphinxATS, an AI-powered Automated Resume Screening and Decision Support System, was introduced in this paper. It models the process of evaluating candidates as a structured multi-attribute optimization problem. The suggested framework converts conventional heuristic resume filtering into a measurable and explicable decision-support procedure by combining Natural Language Processing techniques with semantic similarity analysis and adjustable weighted scoring mechanisms. Transparency in the results of candidate ranking is maintained while systematic processing of unstructured resume data is made possible by the modular system architecture. Comparing the suggested method to traditional keyword-based applicant tracking systems, experimental evaluation shows that it improves semantic alignment and ranking consistency. Automated recruitment workflows are made more interpretable and equitable by incorporating attribute-wise scoring and explainable ranking justification. The framework offers a scalable infrastructure that can adjust to a wide range of job-role requirements and sizable applicant pools, according to systems engineering theory. Future research will concentrate on improving the framework's intelligence and adaptability. In order to guarantee fair candidate evaluation across professional and demographic dimensions, bias mitigation techniques will also be investigated. Real-time scalability and operational deployment in big businesses will be further made possible by integration with cloud-based deployment architectures and enterprise-level HR Management platforms.

ACKNOWLEDGMENT

The authors acknowledge the use of AI-based tools such as ChatGPT and ClaudeAI for grammar refinement and language improvement during the manuscript preparation process. All technical concepts, system design, implementation, and experimental analysis presented in this paper were developed solely by the authors.

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